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Dynamical properties of EMIM-SCN confined in a SiO₂ matrix by means of ¹H NMR relaxometry

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¹H nuclear magnetic resonance relaxometry is applied to investigate the translational and rotational dynamics of ionogels composed of an ionic liquid (IL): 1-ethyl-3-methyl-imidazolium-thiocyanate (EMIM-SCN) confined in a nanoporous SiO₂ matrix. The relaxation studies were performed in the frequency range of 4 kHz–40 MHz and the temperature range of 223–248 K for different concentrations of the IL; the ratio (no. of moles of IL/no. of moles of SiO₂) yields: 1/2, 3/5 and 7/10. A thorough analysis of this large set of experimental data was performed assuming the existence of two fractions of the liquid: a core fraction (near the pore center) and a surface fraction (near the confining walls). It was shown for all concentrations that the confinement does not significantly affect the translational motion near the pore center compared to the dynamics in bulk. The diffusion coefficients in the surface fraction are considerably smaller compared to the core fraction (from one to two orders of magnitude) and the difference becomes larger with increasing temperature. The diffusion coefficients become smaller for higher concentrations – this effect is not large, but visible. Very importantly, it was shown that, despite the interactions with the surface, the diffusion in the surface fraction remains of 3D character. As far as rotational dynamics in the surface fraction is concerned, it slows down compared to the bulk (and the core fraction), but this effect is of the order of factor 2–3.

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1. Introduction

Development of energy storage systems is a vital goal of materials science. As a consequence, the majority of studies are focused on the synthesis and electrochemical characterization of new compounds for electrolytes and electrodes. Nevertheless, there is a growing awareness that further progress in this area depends on a thorough understanding of the mechanisms of ionic transport and interactions with surfaces. In this context transport properties of nano-confined ionic liquids (ILs) have become increasingly more important. An example is development of ionogels where ILs are confined in nanopores of some solid substrates, for instance SiO₂ matrices. Ionogels are attracting increasing interest due to their diverse applications such as in rechargeable batteries, supercapacitors, fuel cells, dye sensitized solar cells, catalysis, separations *etc.*^{1–4} A great challenge in designing ionogels is to be able to preserve as much as possible

the liquid-like ionic mobility despite geometrical restrictions and interactions with the confining walls.

Structural and dynamical properties of confined ILs have been investigated using both theoretical^{5–9} and experimental approaches.^{10–16} Molecular dynamics simulations have led to diverse results (slower or faster ion dynamics) for ILs confined in different host materials.^{6–9} As far as the experimental studies of the ionic dynamics are concerned, one should refer to recent studies of translational diffusion of ILs confined in SiO₂ nanopores using ¹H Magic Angle Spinning (MAS) Nuclear Magnetic Resonance (NMR) spectroscopy,¹³ the Pulsed Field Gradient (PFG)-NMR method,^{11,14,17} and single-frequency NMR relaxation measurements.¹⁷ In the last paper, slowing down of rotational and translational dynamics near pore wall surfaces, and nearly bulk-like dynamics in the center of the pore has been reported for the BMIM cations of 1-butyl-3-methylimidazolium bis-(trifluoromethylsulfonyl)imide (BMIM-TFSI) confined in porous silica. In another study slightly reduced dynamics of diethyl-methylammonium methanesulfonate (DEMA-OMs) confined in nano-porous silica has been reported using PFG-NMR,¹¹ whereas faster dynamics of BMIM-TFSI confined in mesoporous carbon has been indicated by means of quasielastic neutron scattering.¹⁵ More recently, we have investigated the translational and rotational dynamics of 1-butyl-3-methylimidazolium octylsulfate (BMIM-OcSO₄) in bulk and SiO₂ confinement using

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^1H NMR relaxometry.¹⁸ It has been concluded that the translational dynamics of both ions (cation and anion) is very similar – this concerns bulk as well as confined liquids. It has been shown that in the confinement one can clearly distinguish two kinds of translational dynamics – diffusion in the pore center (core-fraction) and diffusion near the pore surface mediated by acts of adsorption to the pore walls. It has been demonstrated that the movement near the surface is described by a correlation time of the order of 10^{-6} s – as a consequence, the effective translational diffusion near the surface is much slower (more than an order of magnitude) than in the bulk. The rotational dynamics of the cation and the anion has been separately revealed. It has been shown that a characteristic time constant (referred to as a rotational correlation time) for the cation is by at least one order of magnitude shorter than for the anion; moreover, it has been shown that the rotational motion is unaffected by the confinement. At the same time an additional motional process of the order of ns, not observed in bulk, has been revealed in confinement.

This shows that NMR relaxometry is a method which gives deep insight into molecular (ionic) dynamics. Standard NMR relaxation experiments are performed at a single resonance frequency (magnetic field). By applying the fast field cycling (FFC) method^{19,20} the resonance frequency can be varied in the range from 4 kHz to 40 MHz (referring to ^1H). As a consequence, by NMR relaxometry one can detect motional processes across a broad range of time scales (from ms to ns) by a single experiment. Most importantly, frequency-dependent relaxation studies possess the exceptional potential to reveal the underlying mechanisms of molecular (ionic) motion. NMR relaxation experiments provide a quantity referred to as spin-lattice relaxation rate, R_1 (*i.e.* a reciprocal spin-lattice relaxation time, $R_1 = T_1^{-1}$). The dependence of the relaxation rate on the resonance frequency, $R_1(\omega)$, is called a relaxation dispersion profile. The predominating origin of ^1H relaxation in liquids is provided by ^1H – ^1H magnetic dipole–dipole interactions that stochastically fluctuate in time due to molecular (ionic) motion. The spin-lattice relaxation rate is given as a linear combination of spectral density functions being the Fourier transform of the corresponding correlation functions for the motion modulating spin interactions causing the relaxation process. Mathematical forms of spectral densities characterizing different kinds of dynamics (translation, rotation) are essentially different – as a consequence, relaxation dispersion experiments unambiguously reveal the mechanism of ionic motion. One should point out that relatively recently, NMR relaxometry has been applied to investigate the translational and rotational dynamics of molecules and ILs.^{21,22} In particular, it has been demonstrated that for free translation motion the diffusion coefficients can be determined from a low frequency slope of the relaxation rate plotted *versus* the square root of the resonance frequency.²³ At the same time, the existence of such a linear dependence proves that the translation motion is of 3D character.

This work is a continuation of our previous studies of the dynamical properties of BMIM-OcSO₄ in SiO₂ confinement.¹⁸ It has two goals: the first one is to take advantage of NMR

relaxometry to get deep insight into the dynamical properties of 1-ethyl-3-methyl-imidazolium-thiocyanate (EMIM-SCN) in a SiO₂ matrix in order to compare the influence of the confinement on the dynamics of the EMIM cations with the confinement effects revealed for BMIM-OcSO₄. The second goal is to explore how the dynamics in confinement depends on the concentration of the IL.

This paper is organized as follows: in Section 2, the theory of ^1H spin-lattice relaxation is presented, Section 3 contains experimental details, while a thorough analysis of the relaxation data for EMIM-SCN in the bulk and SiO₂ confinement and discussion of the results is presented in Section 4. Section 5 includes concluding remarks.

2. Theory

The primary source of ^1H relaxation is magnetic dipole–dipole interactions. For EMIM-SCN one can distinguish two relaxation pathways: intra-ionic and inter-ionic ^1H – ^1H dipolar interactions. As a consequence, the ^1H spin-lattice relaxation rate, $R_1(\omega) = T_1^{-1}(\omega)$ ($\omega = \gamma_{\text{H}}B_0$ denotes the proton resonance frequency in angular frequency units, γ_{H} is the proton gyromagnetic factor) is given as a sum of two terms:^{24,25}

$$R_1(\omega) = R_{1,\text{intra}}(\omega) + R_{1,\text{inter}}(\omega) \quad (1)$$

where $R_{1,\text{intra}}(\omega)$ and $R_{1,\text{inter}}(\omega)$ denote the intra-ionic and inter-ionic ^1H – ^1H relaxation contributions, respectively. Because of the structure of EMIM-SCN the relaxation process involves only cations; SCN anions do not contain protons. The cation–anion ^1H – ^{14}N dipole–dipole interactions can be neglected due to the small gyromagnetic factor of ^{14}N . The intra-ionic (intra-cationic) relaxation rate can be expressed as

$$R_{1,\text{intra}}(\omega) = C_{\text{DD}}[J_{\text{intra}}(\omega) + 4J_{\text{intra}}(2\omega)] \quad (2)$$

where C_{DD} denotes the ^1H – ^1H dipolar relaxation constant. The intra-cationic dipolar interactions are modulated by rotational dynamics of the cation. In the simplest case when the rotational correlation functions are exponential the spectral density $J_{\text{intra}}(\omega)$ is given as:^{26–29}

$$J_{\text{intra}}(\omega) = J_{\text{rot}}(\omega) = \frac{\tau_{\text{rot}}}{1 + \omega^2\tau_{\text{rot}}^2} \quad (3)$$

where τ_{rot} denotes the rotational correlation time. The inter-cationic ^1H – ^1H dipolar interactions fluctuate in time due to relative translational diffusion of the EMIM cations. The corresponding relaxation contribution can be described by an expression similar to eqn (2):^{28–31}

$$R_{1,\text{inter}}(\omega) = \frac{3}{10}N_{\text{H}}\left(\frac{\mu_0}{4\pi}\gamma_{\text{H}}^2\hbar\right)^2[J_{\text{inter}}(\omega) + 4J_{\text{inter}}(2\omega)] \quad (4)$$

where N_{H} is the number of protons per unit volume. Assuming the force-free hard-sphere model of translational diffusion, the spectral density, $J_{\text{inter}}(\omega)$, can be expressed as:^{30,31}

$$J_{\text{inter}}(\omega) = J_{\text{trans}}(\omega) = 72\frac{1}{d^3}\int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \frac{u^2\tau_{\text{trans}}}{u^4 + (\omega\tau_{\text{trans}})^2} du \quad (5)$$

where u denotes an integration variable. The translational correlation time, τ_{trans} , is defined as $\tau_{\text{trans}} = d^2/D_{12}$, where D_{12} denotes a relative translation diffusion coefficient defined as a sum of diffusion coefficients of the interacting species; for identical molecules (ions) one gets $D_{12} = 2D$, while d is referred to as the distance of closest approach for the molecules (ions). The translation diffusion coefficient can also be determined from the low frequency slope of the relaxation rate plotted *versus* square root of the frequency. At low frequencies, when $\omega\tau_{\text{trans}} < 1$, the relaxation rate $R_1(\omega)$ can be approximated by a linear function $R_1(\omega) \cong R_1(0) - B\sqrt{\omega}$.^{32,33} The linear dependence of $R_1(\omega)$ on $\sqrt{\omega}$ observed at low frequencies can be treated as a sign that the molecule (ion) undergoes a three-dimensional diffusion. The coefficient B includes the translation diffusion coefficient and the number of protons per unit volume:^{23,32,33}

$$B = \pi N_{\text{H}} \left(\frac{\mu_0 \gamma_{\text{H}}^2 \hbar}{4\pi} \right)^2 \left(\frac{1 + 4\sqrt{2}}{30} \right) D^{-3/2} \quad (6)$$

It is important to notice that the coefficient B includes, besides the diffusion coefficient and the number of protons per unit volume, N_{H} , only well-known physical constants (it is independent of the distance of closest approach, d).

The dynamics becomes more complex under confinement. One can think about two fractions of the confined liquid: a core (bulk-like) fraction near the center of the pore, associated with relaxation rate $R_1^{\text{core}}(\omega)$ and a fraction near the pore surface (surface fraction), associated with $R_1^{\text{surf}}(\omega)$. Each contribution includes the intra-ionic and inter-ionic relaxation pathways; thus:

$$\begin{aligned} R_1(\omega) &= R_1^{\text{core}}(\omega) + R_1^{\text{surf}}(\omega) = R_{1,\text{intra}}^{\text{core}}(\omega) + R_{1,\text{inter}}^{\text{core}}(\omega) \\ &\quad + R_{1,\text{intra}}^{\text{surf}}(\omega) + R_{1,\text{inter}}^{\text{surf}}(\omega) \\ &= C_{\text{DD}} \left[\frac{\tau_{\text{rot}}^{\text{core}}}{1 + (\omega\tau_{\text{rot}}^{\text{core}})^2} + \frac{4\tau_{\text{rot}}^{\text{core}}}{1 + (2\omega\tau_{\text{rot}}^{\text{core}})^2} \right] \\ &\quad + \frac{108}{5} \left(\frac{\mu_0 \gamma_{\text{H}}^2 \hbar}{4\pi} \right)^2 \frac{N_{\text{H}}^{\text{core}}}{d_{\text{core}}^3} \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \\ &\quad \times \left[\frac{u^2}{u^4 + (\omega\tau_{\text{trans}}^{\text{core}})^2} + \frac{4u^2}{u^4 + (2\omega\tau_{\text{trans}}^{\text{core}})^2} \right] \\ &\quad + C_{\text{DD}} \left[\frac{\tau_{\text{rot}}^{\text{surf}}}{1 + (\omega\tau_{\text{rot}}^{\text{surf}})^2} + \frac{4\tau_{\text{rot}}^{\text{surf}}}{1 + (2\omega\tau_{\text{rot}}^{\text{surf}})^2} \right] \\ &\quad + \frac{108}{5} \left(\frac{\mu_0 \gamma_{\text{H}}^2 \hbar}{4\pi} \right)^2 \frac{N_{\text{H}}^{\text{surf}}}{d_{\text{surf}}^3} \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \\ &\quad \times \left[\frac{u^2}{u^4 + (\omega\tau_{\text{trans}}^{\text{surf}})^2} + \frac{4u^2}{u^4 + (2\omega\tau_{\text{trans}}^{\text{surf}})^2} \right] \end{aligned} \quad (7)$$

The meaning of the parameters used in eqn (7) is as follows: $\tau_{\text{rot}}^{\text{core}}$ and $\tau_{\text{rot}}^{\text{surf}}$ denote the rotational correlation times of EMIM cations in the core fraction and near the surface, respectively, whereas $\tau_{\text{trans}}^{\text{core}}$ and $\tau_{\text{trans}}^{\text{surf}}$ denote the corresponding translational correlation times, defined as $\tau_{\text{trans}}^{\text{core}} = \frac{d_{\text{core}}^2}{2D_{\text{core}}}$ and $\tau_{\text{trans}}^{\text{surf}} = \frac{d_{\text{surf}}^2}{2D_{\text{surf}}}$,

where D^{core} and D^{surf} are the translational diffusion coefficients of the EMIM cations in the core fraction and near the surface, respectively. Analogously, $N_{\text{H}}^{\text{core}}$ and $N_{\text{H}}^{\text{surf}}$ are numbers of protons per unit volume in the core fraction and near the surface, respectively. For liquid in bulk the N_{H} number can be calculated from the simple expression $N_{\text{H}} = nN_{\text{A}}\rho/M$, where M is the molecular mass of the liquid, ρ is its density, N_{A} is the Avogadro number, and n denotes the number of protons per cation. For EMIM-SCN one obtains $N_{\text{H}} = 4.36 \times 10^{-2} \text{ \AA}^{-3}$ (density of EMIM-SCN is 1.11 g cm^{-3}). The value of $N_{\text{H}}^{\text{core}}$ depends on the concentration of the liquid (it is smaller than N_{H}); moreover, one can expect that a part of the ions are adsorbed to the surface; this is reflected by the $N_{\text{H}}^{\text{surf}}$ value.

The concept of the core and surface fractions assumes that the residence lifetime of the ions on the surface is longer than the translational correlation time in bulk. Otherwise, the ions attached to the surface undergo fast exchange with the core fraction and, as a consequence, one cannot distinguish between these fractions – the translational motion of all the ions can be described by a single, effective correlation time. As already mentioned in the Introduction, discussing the translational dynamics near the surface one can distinguish between weak adsorption and strong adsorption cases, *i.e.* the situation when the ions return to the core fraction after a single act of adsorption to the surface (weak adsorption) and the situation when the ions return to the surface many times before escaping to the core fraction (strong adsorption). In the first case the translational diffusion near the surface can be treated as a 3D process, whereas in the second case the diffusion is rather of 2D character. One can distinguish between these cases by analyzing the dependence of the $R_{1,\text{inter}}^{\text{surf}}$ relaxation contribution (after disentangling it from the overall relaxation rate, R_1) on $\sqrt{\omega}$. A linear dependence of $R_{1,\text{inter}}^{\text{surf}}$ on $\sqrt{\omega}$ indicates a 3D mechanism of the translational dynamics. At this stage one should clearly stress that eqn (7) assumes, by the form of the spectral density describing the translational motion near the surface, that the diffusion is of 3D character that implies the weak adsorption case.

3. Experimental details

Silica ionogels were prepared using the non-hydrolytic sol-gel process as described in the literature.^{34,35} For the synthesis of ionogels, the ionic liquid (IL) and formic acid were mixed with TMOS in molar ratio TMOS (1)/HCOOH (8)/IL ($x = 0.5, 0.6$, and 0.7) in a reaction vessel at ambient temperature. Finally, after gelation and aging, stable monolithic ionogels were obtained. These synthesized ionogels have been abbreviated as TMOS (1)/HCOOH (8)/IL ($x = 0.5$): IGL-1; TMOS (1)/HCOOH (8)/IL ($x = 0.6$): IGL-2; TMOS (1)/HCOOH (8)/IL ($x = 0.7$): IGL-3. The ratio (no. of moles of IL/no. of moles of SiO_2) yields: 1/2, 3/5 and 7/10 for IGL-1, IGL-2 and IGL-3, respectively. A typical photograph of the ionogels is given in Fig. 1. The average pore size of the silica ionogels was analyzed by extracting IL from ionogels using a surface area analyzer (Gemini VII 2390t from Micromeritics Instrument

Corporation) at 77 K. The average pore sizes are found to be 7.5, 8.2 and 5.9 nm respectively for the samples IGL-1, IGL-2 and IGL-3, respectively.

^1H spin-lattice relaxation data for EMIM-SCN confined in the SiO_2 matrix have been collected for all the concentrations of the IL. The same temperature range of 223–248 K has been chosen for all the concentrations. The temperatures correspond to those at which ^1H relaxation experiments for EMIM-SCN in bulk have previously been performed.²² The current experiments have been carried out in the frequency range of 4 kHz–40 MHz (for ^1H). The ^1H relaxation data have been collected using a STELAR SPINMASTER FFC2000 Fast Field Cycling (FFC) relaxometer. Below 10 MHz the sample has initially been polarized at a magnetic field corresponding to the ^1H resonance frequency of 24 MHz for a time period set to fourfold of the spin-lattice time at 24 MHz. A recycle delay of fivefold of the spin-lattice time at

40 MHz has been applied. The NMR signal has been acquired with 8 scans for 16 linearly spaced time sets. The following settings have been chosen: the slew rate of the magnetic field – 12 MHz ms^{-1} , the detection frequency – 15.8 MHz , the length of $\pi/2$ pulse – $10\text{ }\mu\text{s}$. Temperature has been controlled by a gas-flow system monitored by the Eurotherm temperature control unit using a feedback control mechanism with input from a thermocouple mounted in the pathway of the air. For high temperature measurements dry air was used, heated prior to entering the sample volume, while for low temperature measurements nitrogen was used. A thermocouple temperature sensor has been inserted into the probehead to ensure accuracy of the temperature measurement of 0.5 K .

It has turned out that the ^1H spin-lattice relaxation is single-exponential for all the samples in the whole frequency and temperature range. An example for 243 K is shown in Fig. 2.



Fig. 1 A typical photograph of an ionogel.

4. Results, analysis and discussion

In Fig. 3 ^1H spin-lattice relaxation data obtained for EMIM-SCN in confinement for different concentrations (IGL-1, IGL-2 and IGL-3) are compared with the data obtained for EMIM-SCN in bulk taken from ref. 22. Two observations can be made at this stage: the relaxation data for EMIM-SCN in confinement considerably differ from the corresponding results for bulk IL and the data for the highest concentration of EMIM-SCN (IGL-3) differ from those for the lower concentrations (IGL-1 and IGL-2). To compare the temperature dependencies of the relaxation data for different concentrations in Fig. 4 of the Appendix, four sets of relaxation data (bulk, IGL-1, IGL-2 and IGL-3) for the temperature range of 223–248 K are shown. One can see that the changes

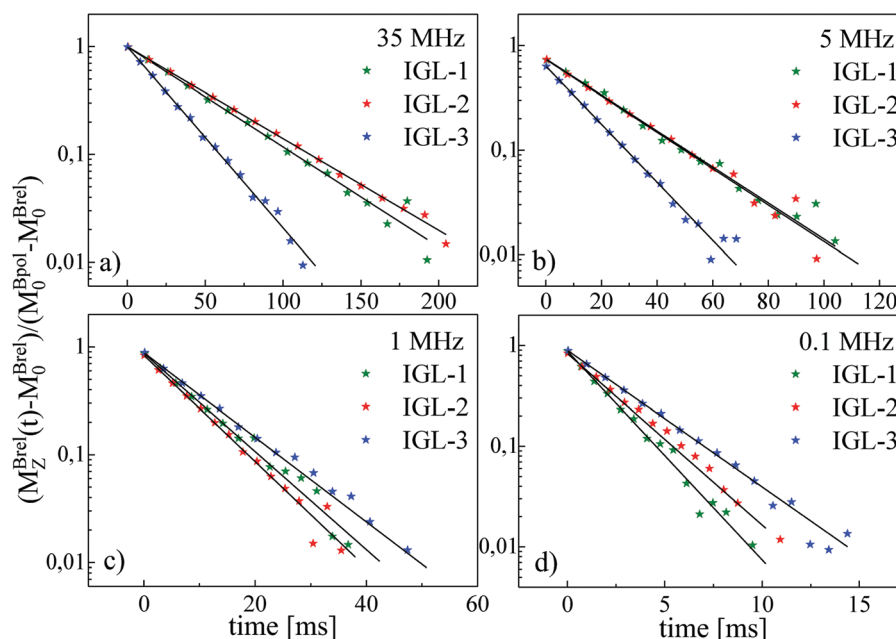


Fig. 2 Time dependence of normalized ^1H magnetization at 243 K for different concentrations of EMIM-SCN confined in SiO_2 matrices (IGL-1, IGL-2 and IGL-3) at (a) 35 MHz, (b) 5 MHz, (c) 1 MHz, and (d) 0.1 MHz. Solid lines – single exponential fits.

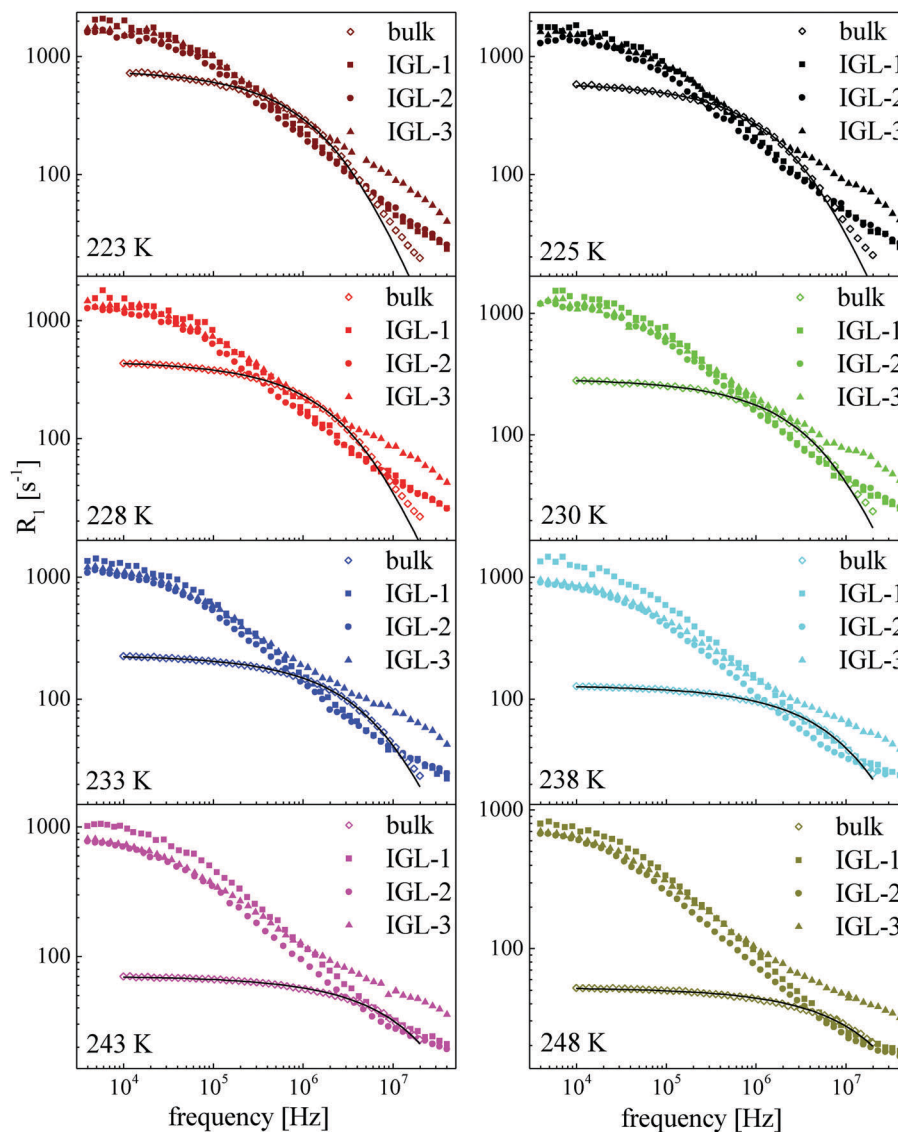


Fig. 3 ^1H spin–lattice relaxation data for EMIM-SCN (different concentrations) in a SiO_2 matrix compared with relaxation data for EMIM-SCN in bulk taken from ref. 21; solid lines – fits to the relaxation data for EMIM-SCN in bulk.²¹

in the relaxation rates with temperature are less pronounced for the confined liquid than for bulk.

One should explain at this stage that the relaxation of EMIM-SCN in bulk²² is dominated by the translational (inter-cationic) contribution. At low temperatures and high frequencies one observes some discrepancies between the experimental data and the theoretical lines representing the translational contribution (see Fig. 3 and 4a). These discrepancies can be attributed to the rotational dynamics in bulk. Before proceeding with the analysis of the dynamics of EMIM-SCN in confinement, we have re-analyzed the relaxation data for bulk at 220 K (ref. 22 also contains data for 220 K, 223 K, 225 K and 228 K, including a rotational (intra-cationic) contribution, as shown in Fig. 5. This has given us the value of the dipolar relaxation constant, $C_{\text{DD}} = 1.14 \times 10^9 \text{ Hz}^2$ (the value is somewhat smaller than for BMIM¹⁸ as expected) and the rotational correlation times for EMIM-SCN in bulk: $\tau_{\text{rot}} = 2.77 \times 10^{-9} \text{ s}$ (220 K), $2.60 \times 10^{-9} \text{ s}$ (223 K),

$2.26 \times 10^{-9} \text{ s}$ (225 K) and $2.16 \times 10^{-9} \text{ s}$ (228 K). Actually only at 220 K the relaxation contribution representing rotation of EMIM cations in bulk shows a slight dispersion that allows C_{DD} to be determined. At higher temperatures the motion becomes too fast to lead to a frequency-dependent contribution, but knowing the C_{DD} value one can still obtain the rotational correlation time from the frequency-independent term; above 228 K the contribution becomes negligible.

We begin the analysis with the intermediate concentration of IL-confined sample: IGL-2. Discussing the relative timescales of the motional processes leading to the ^1H relaxation of the confined liquid, one should take into account that according to the Stokes–Einstein equation: $\tau_{\text{trans}}/\tau_{\text{rot}} = 9$; it has been shown that for “real” liquids the ratio is even larger – about 30–40.³⁶ Moreover, it is reasonable to expect that for the confined liquid the translational dynamics near the surface is slower

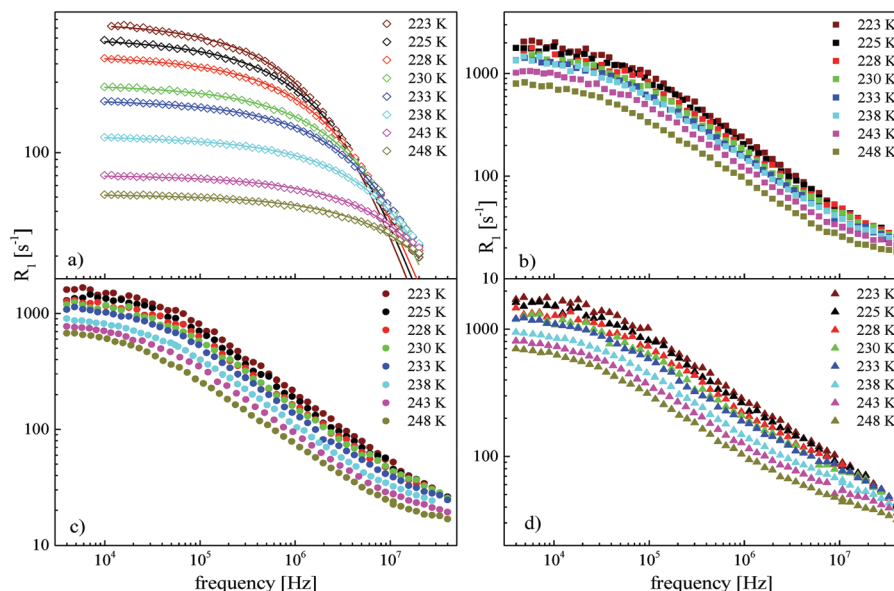


Fig. 4 ^1H spin-lattice relaxation dispersion profiles for EMIM-SCN in (a) bulk (taken from ref. 21) and in a SiO_2 matrix – (b) IGL-1, (c) IGL-2, and (d) IGL-3 concentration of the IL in the temperature range of 220–248 K. Solid lines – fits of the relaxation data for EMIM-SCN in bulk.²¹

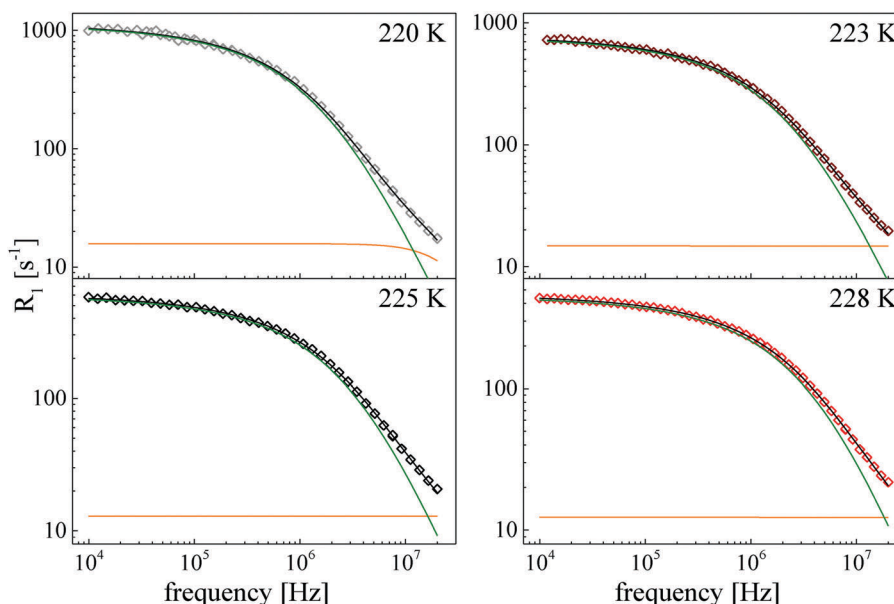


Fig. 5 ^1H spin-lattice relaxation profiles of EMIM-SCN in bulk at 220 K, 223 K, 225 K and 228 K,²¹ decomposed into the contributions described by eqn (1): green lines – inter-cationic contributions, orange lines – intra-cationic contributions, black lines – total relaxation rates (R_1^{bulk}).

than in the core fraction. As a consequence, one can conclude that the relaxation in the low frequency range is dominated by the translational dynamics near the surface (the slowest process), the intermediate frequency range reflects the translational dynamics of the core fraction and rotational dynamics near the surface, and the rotational dynamics of the core fraction manifests itself at high frequencies. Thus, the relaxation data in the intermediate and high frequency ranges can be reproduced in terms of the following contributions:

$$\begin{aligned}
 R_1(\omega) &= R_{1,\text{inter}}^{\text{core}}(\omega) + R_{1,\text{intra}}^{\text{surf}}(\omega) + A \\
 &= \frac{108}{5} \left(\frac{\mu_0 \gamma_{\text{H}}^2 \hbar}{4\pi} \right)^2 \frac{N_{\text{H}}^{\text{core}}}{d^3} \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \\
 &\quad \times \left[\frac{u^2}{u^4 + (\omega \tau_{\text{trans}}^{\text{core}})^2} + \frac{4u^2}{u^4 + (2\omega \tau_{\text{trans}}^{\text{core}})^2} \right] \\
 &\quad + C_{\text{DD}} \left[\frac{\tau_{\text{rot}}^{\text{surf}}}{1 + (\omega \tau_{\text{rot}}^{\text{surf}})^2} + \frac{4\tau_{\text{rot}}^{\text{surf}}}{1 + (2\omega \tau_{\text{rot}}^{\text{surf}})^2} \right] + A
 \end{aligned} \quad (8)$$

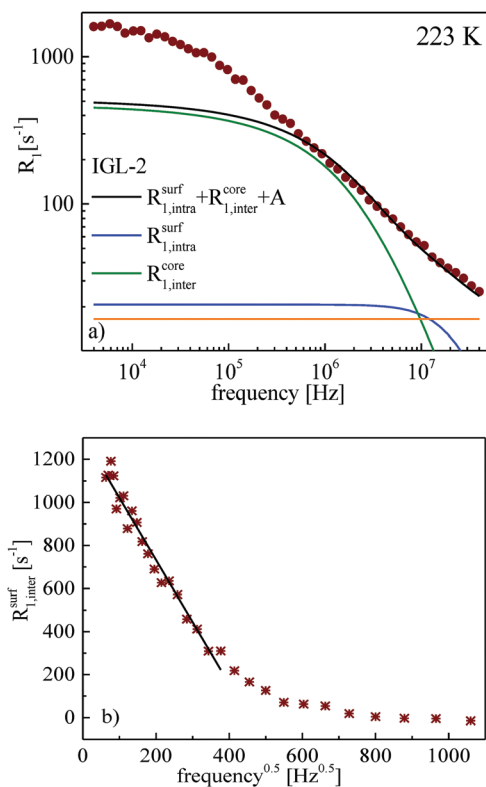


Fig. 6 (a) Contributions to the overall ^1H spin–lattice relaxation of EMIM-SCN in SiO_2 (IGL-2) dominating the relaxation rate at intermediate and high frequencies and their sum; (b) difference between the overall relaxation rate and the sum of the contributions shown in (a) plotted versus $\sqrt{\omega/2\pi}$; black line – linear fit in the low frequency range.

as shown in Fig. 6a for IGL-2 at 223 K. The equation contains four adjustable parameters: d_{core} , $\tau_{\text{trans}}^{\text{core}}$, $\tau_{\text{rot}}^{\text{surf}}$ and A . One could expect that the parameter $A = 5C_{\text{DD}}\tau_{\text{rot}}^{\text{core}}$ corresponds to the first term of eqn (7), $R_{1,\text{intra}}^{\text{core}}$, when the dynamics is so fast that the condition $\omega\tau_{\text{rot}}^{\text{core}} \ll 1$ is fulfilled (then the contribution becomes frequency independent). In this case the term can also be considered as known from the analysis of the data for EMIM-SCN in bulk, but, anticipating the results, we have decided to keep it adjustable. The number of protons per unit volume for EMIM-SCN in bulk yields $N_{\text{H}} = 4.36 \times 10^{-2} \text{ \AA}^{-3}$; the $N_{\text{H}}^{\text{core}}$ value can be obtained knowing the molecular masses and densities of EMIM-SCN and SiO_2 and the molar ratio of

these components (3/5 for IGL-2); one obtains $N_{\text{H}}^{\text{core}} = 3.34 \times 10^{-2} \text{ \AA}^{-3}$, while the value of $C_{\text{DD}} = 1.14 \times 10^9 \text{ Hz}^2$ is known from the analysis of the relaxation data for bulk. Moreover, one can expect that the d_{core} and $\tau_{\text{trans}}^{\text{core}}$ values should be close to those obtained in ref. 22 for EMIM-SCN in bulk at 223 K: $d = 2.45 \text{ \AA}$ and $\tau_{\text{trans}} = 1.67 \times 10^{-7} \text{ s}$. Therefore, performing the fitting we have allowed only for small deviations from these values. The fitting has converged to $d_{\text{core}} = 2.7 \text{ \AA}$, $\tau_{\text{trans}}^{\text{core}} = 1.67 \times 10^{-7} \text{ s}$. The individual contributions to the relaxation rate, $R_{1,\text{inter}}^{\text{core}}$, $R_{1,\text{intra}}^{\text{surf}}$ and A as well as a sum of these terms are shown in Fig. 6a. In the next step the difference between the experimental data and the theoretical values of the $(R_{1,\text{inter}}^{\text{core}} + R_{1,\text{intra}}^{\text{surf}} + A)$ has been taken and plotted versus the square root of the resonance frequency (Fig. 6b). The difference corresponds to the $R_{1,\text{inter}}^{\text{surf}}$ contribution. One can see from Fig. 6b that at low frequencies it follows a linear dependence on $\sqrt{\omega}$ which proves that the translational diffusion of EMIM cations near the surface is indeed of 3D character. The $R_{1,\text{inter}}^{\text{surf}}$ contribution is described by the last term of eqn (7). The expression for $R_{1,\text{inter}}^{\text{surf}}$ contains three unknown parameters d_{surf} , $\tau_{\text{trans}}^{\text{surf}}$ and $N_{\text{H}}^{\text{surf}}$. The last value can differ from $N_{\text{H}}^{\text{core}}$ because a fraction of the cations can be adhered to the pore walls. The values of d_{surf} , $\tau_{\text{trans}}^{\text{surf}}$ and $N_{\text{H}}^{\text{surf}}$ cannot be unambiguously determined from the expression of eqn (7), but a relationship between $N_{\text{H}}^{\text{surf}}$ and D^{surf} can be determined from the low frequency slope of $R_{1,\text{inter}}^{\text{surf}}$ versus $\sqrt{\omega}$ by means of eqn (6); it yields $N_{\text{H}}^{\text{surf}}(D^{\text{surf}})^{-3/2} = 2.9 \times 10^{48} \text{ m}^{-6} \text{ s}^{3/2}$. On the basis of this preliminary analysis, a fitting of the overall relaxation dispersion data in the whole frequency range has been performed with the following parameters: d_{core} , $\tau_{\text{trans}}^{\text{core}}$, $\tau_{\text{rot}}^{\text{surf}}$, A , d_{surf} and $N_{\text{H}}^{\text{surf}}$. The first four parameters have been set to the values obtained from the partial fitting and only a small refine-

ment of them has been allowed; the parameter $\tau_{\text{trans}}^{\text{surf}} = \frac{d_{\text{surf}}^2}{D^{\text{surf}}}$ has been eliminated as a result of the relationship between $N_{\text{H}}^{\text{surf}}$ and D^{surf} . The obtained parameters are collected in Table 1. Although the parameters somewhat differ from their initial values – for instance $\tau_{\text{trans}}^{\text{core}}$ now yields $2.10 \times 10^{-7} \text{ s}$ instead of the value for bulk: $1.67 \times 10^{-7} \text{ s}$, the differences are small, so the assumption that the dynamics of the core fraction is bulk-like is confirmed. The value of the correlation time characterizing the rotational dynamics near the surface yields $\tau_{\text{rot}}^{\text{surf}} = 6.0 \times 10^{-9} \text{ s}$; this means that the rotational dynamics near the surface are somewhat slower than in bulk. The frequency-independent term yields

Table 1 Parameters describing translational and rotational dynamics of EMIM-SCN in a SiO_2 matrix – IGL-2: ($C_{\text{DD}} = 1.14 \times 10^9 \text{ Hz}^2$, $N_{\text{H}}^{\text{core}} = 3.34 \times 10^{-2} \text{ \AA}^{-3}$, $N_{\text{H}}^{\text{surf}} = 3.68 \times 10^{-3} \text{ \AA}^{-3}$); values of the diffusion coefficients, D^{core} and D^{surf} , have been calculated from the relationships $\tau_{\text{trans}}^{\text{core}} = \frac{d_{\text{core}}^2}{2D^{\text{core}}}$ and $\tau_{\text{trans}}^{\text{surf}} = \frac{d_{\text{surf}}^2}{2D^{\text{surf}}}$, respectively

T [K]	$\tau_{\text{rot}}^{\text{surf}}$ [s]	d_{core} [Å]	$\tau_{\text{trans}}^{\text{core}}$ [s]	d_{surf} [Å]	$\tau_{\text{trans}}^{\text{core}}$ [s]	A [s $^{-1}$]	D^{core} [m 2 s $^{-1}$]	D^{surf} [m 2 s $^{-1}$]	Rel. err. [%]
223	6.0×10^{-9}	2.93	2.04×10^{-7}	2.25	2.81×10^{-6}	17.5	2.10×10^{-13}	0.90×10^{-14}	5.9
225	5.6×10^{-9}	2.98	1.62×10^{-7}	2.25	2.41×10^{-6}	16.9	2.75×10^{-13}	1.05×10^{-14}	6.6
228	4.0×10^{-9}	3.04	1.29×10^{-7}	2.25	2.37×10^{-6}	15.0	3.58×10^{-13}	1.07×10^{-14}	10.0
230	3.3×10^{-9}	3.04	0.98×10^{-7}	2.25	2.35×10^{-6}	14.8	4.70×10^{-13}	1.08×10^{-14}	5.7
233	2.7×10^{-9}	3.04	0.65×10^{-7}	2.25	2.50×10^{-6}	12.3	7.05×10^{-13}	1.13×10^{-14}	4.2
238		3.04	0.51×10^{-7}	2.30	1.94×10^{-6}	19.4 (A')	9.10×10^{-13}	1.37×10^{-14}	11.4
243		3.04	0.31×10^{-7}	2.35	1.78×10^{-6}	16.7 (A')	1.47×10^{-12}	1.55×10^{-14}	5.1
248		3.04	0.24×10^{-7}	2.45	1.72×10^{-6}	12.3 (A')	1.94×10^{-12}	1.75×10^{-14}	8.5

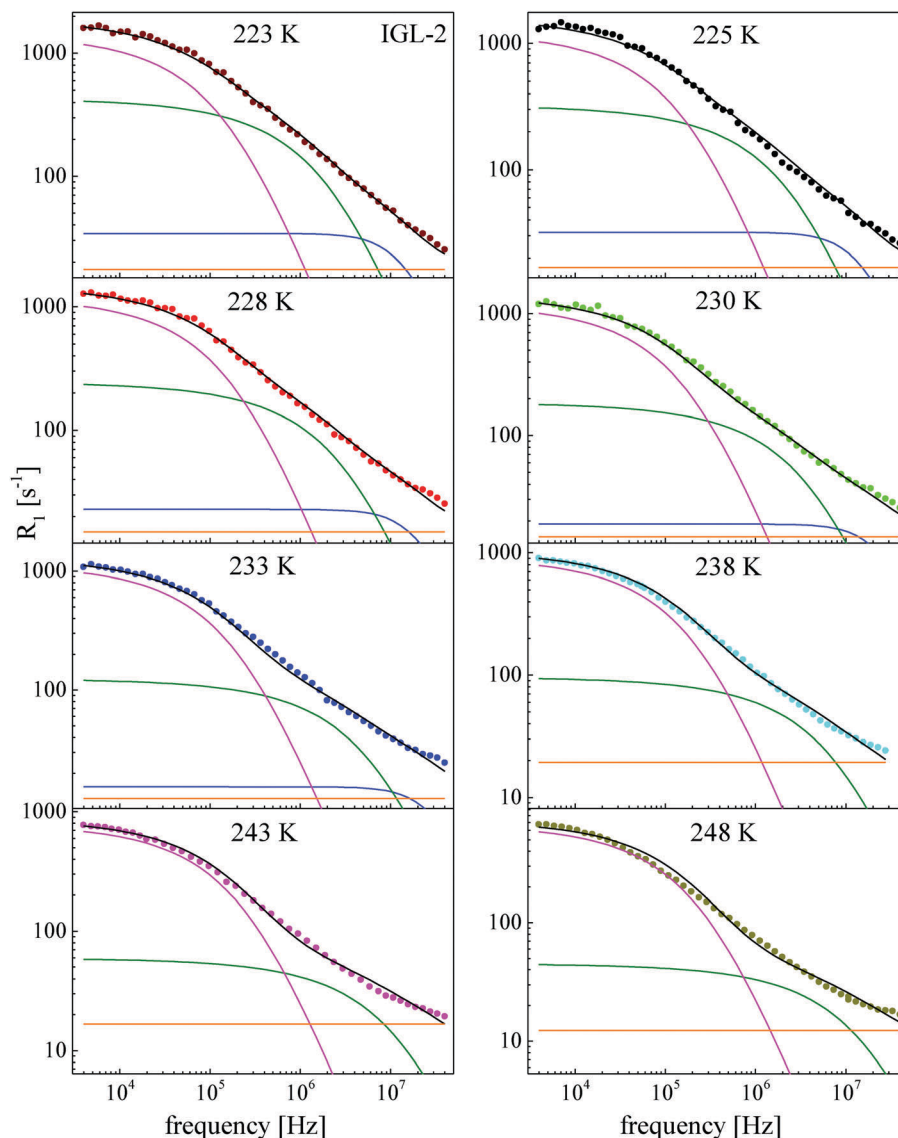


Fig. 7 ^1H spin-lattice relaxation data for EMIM-SCN in a SiO_2 matrix (IGL-2); red lines – $R_1(\omega)$ decomposed into: $R_{1,\text{inter}}^{\text{core}}(\omega)$ – green lines, $R_{1,\text{inter}}^{\text{surf}}(\omega)$ – magenta lines, $R_{1,\text{intra}}^{\text{surf}}(\omega)$ – blue lines, A (A') – orange lines.

$A = 17.5 \text{ s}^{-1}$ – it is somewhat larger than for EMIM-SCN in bulk ($A = 14.8 \text{ s}^{-1}$). This can suggest two possibilities: the rotational dynamics of EMIM cations in the core fraction is also somewhat slower than in bulk or/and the dynamics of the methyl group (in the EMIM cation) has become slower in the confinement leading to a noticeable contribution. The result of the fitting is shown in Fig. 7. The relaxation data up to 233 K have been fitted profiting from the parameters obtained from the analysis performed for 223 K: $N_{\text{H}}^{\text{core}} = 3.34 \times 10^{-2} \text{ \AA}^{-3}$, $N_{\text{H}}^{\text{surf}} = 3.68 \times 10^{-3} \text{ \AA}^{-3}$ and $C_{\text{DD}} = 1.14 \times 10^9 \text{ Hz}^2$ (these parameters have been fixed) and using six adjustable parameters $\tau_{\text{rot}}^{\text{surf}}$, d_{core} , $\tau_{\text{trans}}^{\text{core}}$, d_{surf} , $\tau_{\text{trans}}^{\text{core}}$ and A . For 238 K and above, the rotational dynamics of EMIM cations near the surface becomes too fast to lead to a frequency-dependent contribution. Therefore, it has been included into the frequency-independent term denoted now as $A' = A + 5C_{\text{DD}}\tau_{\text{rot}}^{\text{surf}}$, where $5C_{\text{DD}}\tau_{\text{rot}}^{\text{surf}}$ is the limiting value of the second term in eqn (8)

under the condition $\omega\tau_{\text{rot}}^{\text{surf}} \ll 1$. As a consequence the fits have been performed with five adjustable parameters $\tau_{\text{rot}}^{\text{surf}}$, d_{core} , $\tau_{\text{trans}}^{\text{core}}$, d_{surf} , $\tau_{\text{trans}}^{\text{core}}$ and A' . The outcome of the analysis is shown in Fig. 8, whereas the obtained parameters are collected in Table 1. Fig. 7 shows how the diffusion coefficients D^{core} and D^{surf} change with reciprocal temperature; D^{core} has been compared with diffusion coefficients for EMIM-SCN in bulk obtained in ref. 22. Diffusion near the surface shows a weaker dependence on temperature than diffusion in the core fraction – D^{surf} changes in the temperature range of 223–248 K by about factor 2, whereas D^{core} changes by about one order of magnitude. This is reflected by activation energies $39.2 \pm 1.6 \text{ kJ mol}^{-1}$ and $11.7 \pm 0.8 \text{ kJ mol}^{-1}$ for translational diffusion near the surface and in the core fraction, respectively, determined from the Arrhenius formula.^{37,38} At lower temperatures the diffusion coefficient D^{core} is very close to the diffusion coefficient in bulk, at higher

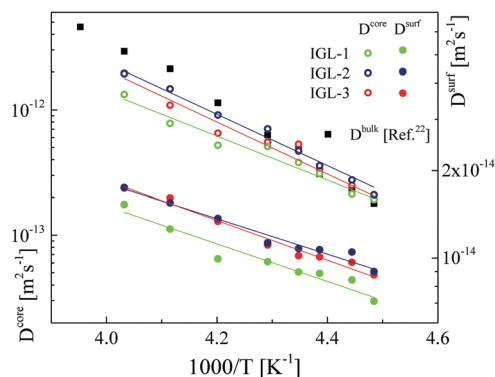


Fig. 8 Diffusion coefficients versus reciprocal of temperature. Solid lines: fits according to the Arrhenius formula, the activation energies obtained for the core fraction are: IGL-1 – 33.6 ± 2.5 kJ mol⁻¹, IGL-2 – 39.2 ± 1.6 kJ mol⁻¹, IGL-3 – 39.7 ± 2.9 kJ mol⁻¹, while for the surface fraction they yield: IGL-1 – 12.4 ± 1.3 kJ mol⁻¹, IGL-2 – 11.7 ± 0.8 kJ mol⁻¹, IGL-3 – 13.1 ± 0.7 kJ mol⁻¹.

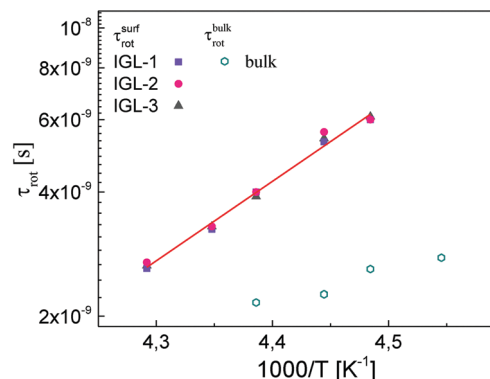


Fig. 9 Rotational correlation times near the surface of SiO₂ pores compared with the rotational correlation time for EMIM-SCN in bulk. Solid line: concatenated fit of $\tau_{\text{rot}}^{\text{surf}}$ according to the Arrhenius formula, activation energy: 37.0 ± 1.3 kJ mol⁻¹.

temperatures the deviations are more significant, but they do not exceed factor 2. Fig. 9 shows the temperature dependence of the correlation time $\tau_{\text{rot}}^{\text{surf}}$, describing rotational dynamics of EMIM cations in the surface fraction. For comparison values of rotational correlation time for EMIM-SCN in bulk are shown.

Analogous analysis has been performed for the lower concentration of EMIM-SCN, IGL-1. The numbers of protons

per unit volume, $N_{\text{H}}^{\text{core}}$ and $N_{\text{H}}^{\text{surf}}$, have been scaled proportionally to the concentration yielding $N_{\text{H}}^{\text{core}} = 3.19 \times 10^{-2} \text{ \AA}^{-3}$, and $N_{\text{H}}^{\text{surf}} = 3.51 \times 10^{-3} \text{ \AA}^{-3}$, respectively. Thus, the parameters $\tau_{\text{rot}}^{\text{surf}}$, d_{core} , $\tau_{\text{trans}}^{\text{core}}$, d_{surf} , $\tau_{\text{trans}}^{\text{core}}$ and A have been treated as adjustable up to 238 K, while above this temperature, for the same reason as in the case of IGL-2 concentration, the set of parameters has been reduced to $\tau_{\text{rot}}^{\text{surf}}$, d_{core} , $\tau_{\text{trans}}^{\text{core}}$, d_{surf} , $\tau_{\text{trans}}^{\text{core}}$ and A' . As the relaxation data for both concentrations (IGL-2 and IGL-1) are similar, the adjustable parameters d_{core} , $\tau_{\text{trans}}^{\text{core}}$, d_{surf} , $\tau_{\text{rot}}^{\text{surf}}$, A have converged to values close to those obtained for IGL-2. They are listed in Table 2 and the fits are shown in Fig. 10.

As already pointed out, the relaxation results for sample IGL-3 (the highest concentration of EMIM-SCN in the SiO₂ matrix) considerably differ in the intermediate and high frequency ranges from those for lower concentrations. Therefore, for this concentration, the data analysis has been performed including one more parameter – the dipolar relaxation constant, C_{DD} . The number of protons per unit volume, $N_{\text{H}}^{\text{core}} = 3.45 \times 10^{-2} \text{ \AA}^{-3}$, and $N_{\text{H}}^{\text{surf}} = 3.80 \times 10^{-3} \text{ \AA}^{-3}$, have been calculated as done previously, assuming their proportionality to the IL concentration. Thus, the analysis up to 233 K has been performed in terms of the following parameters C_{DD} , $\tau_{\text{rot}}^{\text{surf}}$, d_{core} , $\tau_{\text{trans}}^{\text{core}}$, d_{surf} , $\tau_{\text{trans}}^{\text{core}}$ and A , while for the three highest temperatures, 238 K, 243 K and 248 K, this set of parameters has been replaced by $\tau_{\text{rot}}^{\text{surf}}$, d_{core} , $\tau_{\text{trans}}^{\text{core}}$, d_{surf} , $\tau_{\text{trans}}^{\text{core}}$ and A' , because the relaxation contribution associated with EMIM rotation in the surface fraction has become frequency independent as a result of its short correlation time. The analysis is shown in Fig. 11, the obtained parameters are collected in Table 3, the diffusion coefficients D^{core} and D^{surf} and the rotational correlation time $\tau_{\text{rot}}^{\text{surf}}$ are plotted versus reciprocal temperature in Fig. 8 and 9, respectively. Both the translational and the rotational parameters are close to those obtained for the lower concentrations of EMIM-SCN. The difference in the relaxation rates at intermediate and high frequencies is reflected by the dipolar relaxation constant, C_{DD} , which is now by about factor 2 larger than for bulk and the lower concentrations. One should also notice that the frequency-independent time (attributed to rotational dynamics in the core fraction and hence dependent on C_{DD}) is also larger by about factor 2 compared to the concentrations of IL confined in samples IGL-1 and IGL-2. The larger dipolar relaxation constant suggests possible clustering effects caused by restricted space combined with high concentration of the liquid.

Table 2 Parameters describing translational and rotational dynamics of EMIM-SCN in a SiO₂ matrix – IGL-1: ($C_{\text{DD}} = 1.14 \times 10^9$ Hz², $N_{\text{H}}^{\text{core}} = 3.19 \times 10^{-2} \text{ \AA}^{-3}$, $N_{\text{H}}^{\text{surf}} = 3.51 \times 10^{-3} \text{ \AA}^{-3}$); values of the diffusion coefficients, D^{core} and D^{surf} , have been calculated in the same way as in Table 1

T [K]	$\tau_{\text{rot}}^{\text{surf}}$ [s]	d_{core} [Å]	$\tau_{\text{trans}}^{\text{core}}$ [s]	d_{surf} [Å]	$\tau_{\text{trans}}^{\text{core}}$ [s]	A [s ⁻¹]	D^{core} [m ² s ⁻¹]	D^{surf} [m ² s ⁻¹]	Rel. err. [%]
223	6.0×10^{-9}	2.86	2.13×10^{-7}	1.99	2.78×10^{-6}	17.5	1.92×10^{-13}	7.11×10^{-15}	6.0
225	5.3×10^{-9}	2.92	2.00×10^{-7}	2.04	2.47×10^{-6}	16.9	2.13×10^{-13}	8.42×10^{-15}	5.7
228	4.0×10^{-9}	2.94	1.40×10^{-7}	2.04	2.35×10^{-6}	17.2	3.09×10^{-13}	8.86×10^{-15}	7.8
230	3.3×10^{-9}	2.95	1.15×10^{-7}	2.04	2.32×10^{-6}	15.8	3.80×10^{-13}	8.97×10^{-15}	3.7
233	2.6×10^{-9}	2.97	0.87×10^{-7}	2.04	2.14×10^{-6}	12.1	5.09×10^{-13}	9.73×10^{-15}	3.1
238		2.97	0.85×10^{-7}	2.04	2.1×10^{-6}	22.7 (A')	5.22×10^{-13}	9.95×10^{-15}	6.3
243		3.00	0.58×10^{-7}	2.09	1.74×10^{-6}	18.0 (A')	7.80×10^{-13}	1.26×10^{-14}	4.3
248		3.07	0.35×10^{-7}	2.28	1.83×10^{-6}	14.3 (A')	1.33×10^{-12}	1.53×10^{-14}	4.6

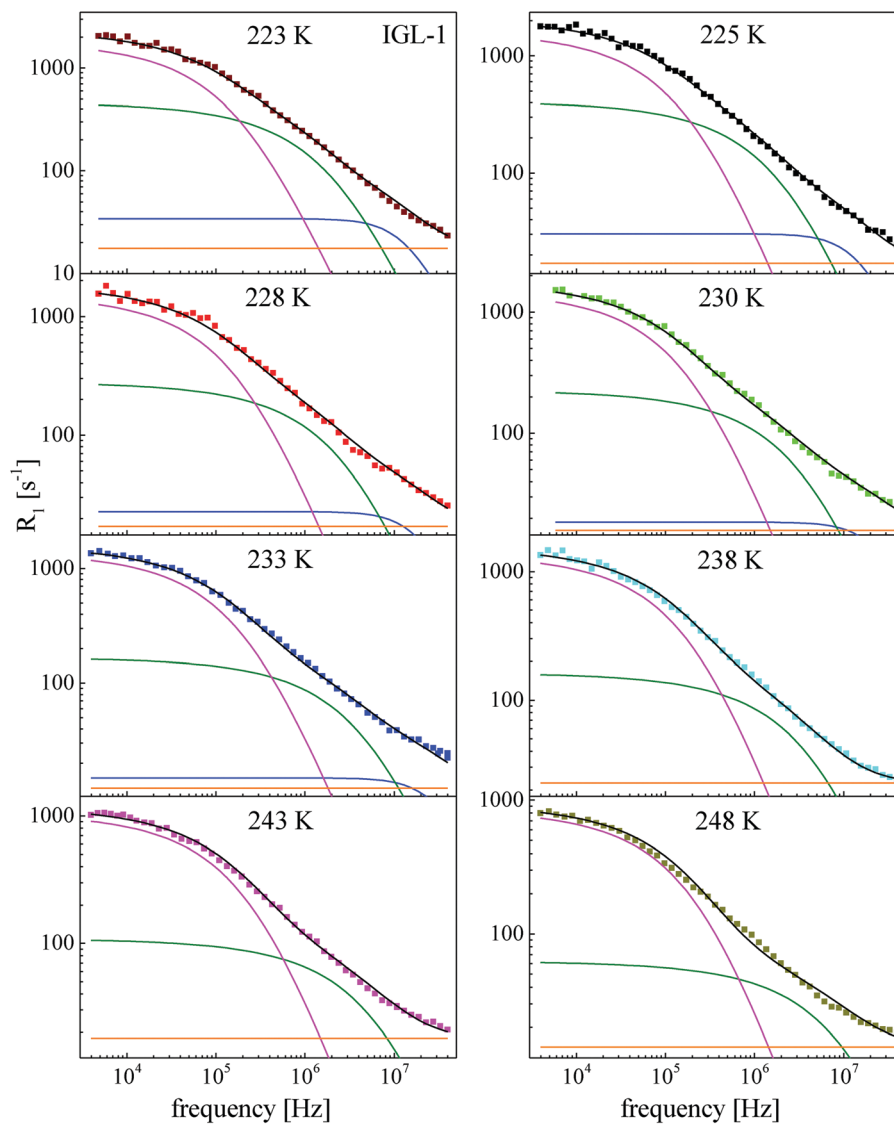


Fig. 10 ^1H spin-lattice relaxation data for EMIM-SCN in a SiO_2 matrix (IGL-1); red lines – $R_1(\omega)$ decomposed into: $R_{1,\text{inter}}^{\text{core}}(\omega)$ – green lines, $R_{1,\text{inter}}^{\text{surf}}(\omega)$ – magenta lines, $R_{1,\text{intra}}^{\text{surf}}(\omega)$ – blue lines, A (A') – orange lines.

5. Conclusions

^1H spin-lattice NMR relaxation studies covering the frequency range of 4 kHz–40 MHz have been performed for ionogels composed of EMIM-SCN in a SiO_2 matrix. The studies have been carried out in the temperature range of 223–248 K. It has been demonstrated that the dynamical properties of the ionogels are related to the existence of two fractions of the IL: a core fraction (close to the pore centers) and a surface fraction (close to the confining walls). The relaxation data have been analyzed in terms of a theoretical model including four contributions to the overall relaxation rate: inter-cation and intra-cation terms (associated with translational and rotational dynamics, respectively) for both fractions. The analysis has revealed that translational diffusion in the core fraction is, in the considered temperature range, similar to the diffusion of EMIM cations in EMIM-SCN in bulk (the results obtained for the ionogels

have been compared with those for EMIM-SCN in bulk²²). However, with increasing temperature one can see the tendency of the translational dynamics in the core fraction to get slower compared to the diffusion in bulk. This is not surprising taking into account that for faster diffusion the geometrical restrictions become more relevant. The translational diffusion coefficient in the core fraction is only very weakly dependent on the concentration of the IL and it ranges between about $10^{-13} \text{ m}^2 \text{ s}^{-1}$ (223 K) and $10^{-12} \text{ m}^2 \text{ s}^{-1}$ (248 K). Rotational dynamics in the core fraction is too fast to lead to a relaxation dispersion as in the case of EMIM-SCN in bulk. This shows that the confinement does not significantly affect the rotational motion near the pore centers. As far as translational dynamics near the surface is concerned, the very important conclusion is that despite the repetitive acts of adsorption to the surface the translational dynamics is of 3D character. The diffusion coefficients for the surface fraction are at the low temperature (223 K) by about an

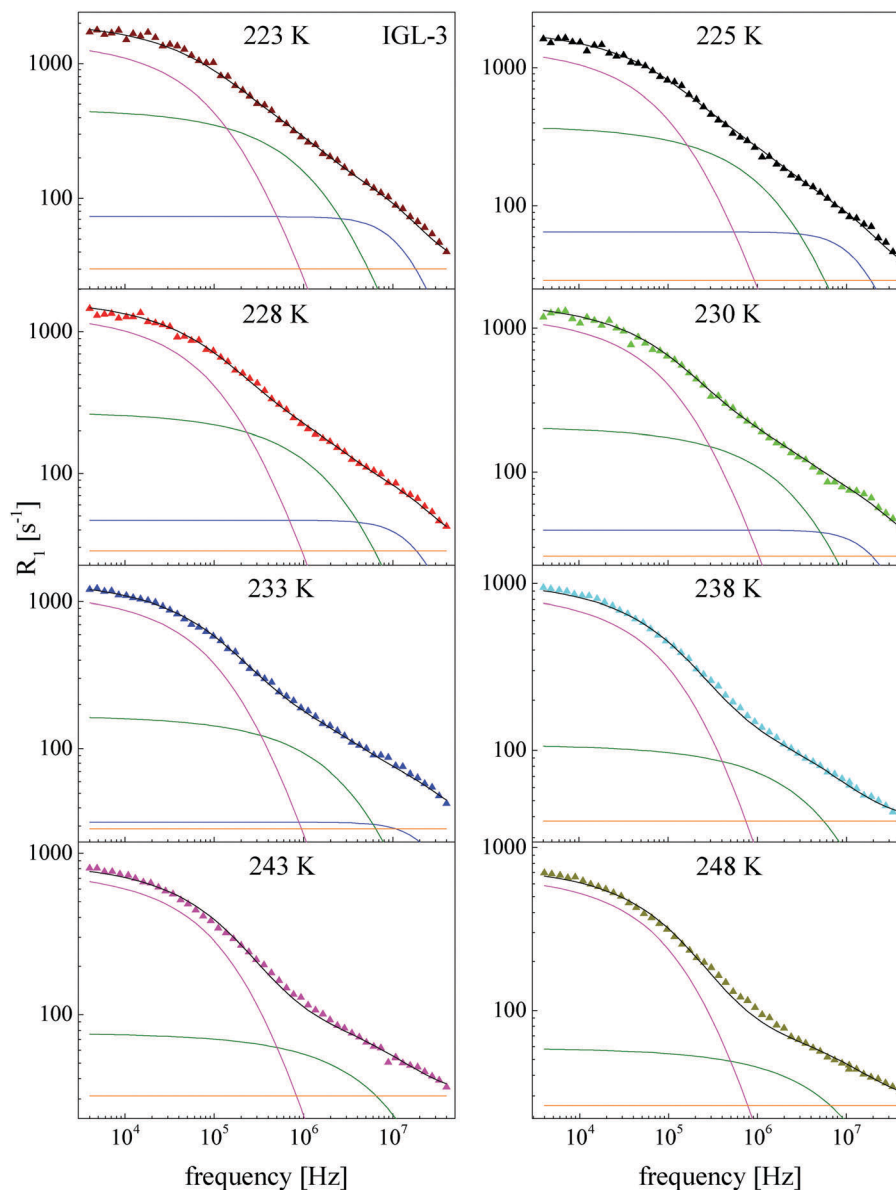


Fig. 11 ^1H spin–lattice relaxation data for EMIM-SCN in a SiO_2 matrix (IGL-3); red lines – $R_1(\omega)$ decomposed into: $R_{1,\text{inter}}^{\text{core}}(\omega)$ – green lines, $R_{1,\text{inter}}^{\text{surf}}(\omega)$ – magenta lines, $R_{1,\text{intra}}^{\text{surf}}(\omega)$ – blue lines, A (A') – orange lines.

Table 3 Parameters describing translational and rotational dynamics of EMIM-SCN in a SiO_2 matrix – IGL-3; ($C_{\text{DD}} = 2.4 \times 10^9 \text{ Hz}^2$, $N_{\text{H}}^{\text{core}} = 3.45 \times 10^{-2} \text{ \AA}^{-3}$, $N_{\text{H}}^{\text{surf}} = 3.80 \times 10^{-3} \text{ \AA}^{-3}$); values of the diffusion coefficients, D^{core} and D^{surf} , have been calculated in the same way as in Table 1

T [K]	$\tau_{\text{rot}}^{\text{surf}}$ [s]	d_{core} [Å]	$\tau_{\text{trans}}^{\text{core}}$ [s]	d_{surf} [Å]	$\tau_{\text{trans}}^{\text{surf}}$ [s]	A [s^{-1}]	D^{core} [$\text{m}^2 \text{ s}^{-1}$]	D^{surf} [$\text{m}^2 \text{ s}^{-1}$]	Rel. err. [%]
223	6.1×10^{-9}	2.87	1.99×10^{-7}	2.18	2.59×10^{-6}	30.1	1.96×10^{-13}	8.76×10^{-15}	5.8
225	5.4×10^{-9}	2.87	1.64×10^{-7}	2.18	2.52×10^{-6}	29.1	2.50×10^{-13}	9.68×10^{-15}	5.0
228	3.9×10^{-9}	2.87	1.16×10^{-7}	2.18	2.40×10^{-6}	28.4	3.37×10^{-13}	1.01×10^{-14}	4.7
230	3.3×10^{-9}	2.81	8.38×10^{-8}	2.18	2.20×10^{-6}	26.2	5.32×10^{-13}	1.02×10^{-14}	5.7
233	2.7×10^{-9}	2.87	7.20×10^{-8}	2.24	2.17×10^{-6}	28.8	5.52×10^{-13}	1.11×10^{-14}	2.8
238		2.61	3.47×10^{-8}	2.35	1.94×10^{-6}	37.7 (A')	6.52×10^{-13}	1.34×10^{-14}	2.7
243		2.53	2.27×10^{-8}	2.40	1.79×10^{-6}	31.3 (A')	1.09×10^{-12}	1.61×10^{-14}	4.8
248		2.61	1.89×10^{-8}	2.58	1.77×10^{-6}	26.0 (A')	1.96×10^{-12}	1.74×10^{-14}	6.8

order of magnitude smaller compared to the core fraction; the difference becomes larger at higher temperatures – at 248 K it reaches two orders of magnitude. The diffusion coefficients near

the surface do not change much with temperature – by about factor 2. Although the concentration does not influence the diffusion coefficients to a large extent, one can see that the

dynamics becomes somewhat faster with increasing concentration. This effect could be explained by “surface accessibility.” The diffusion gets slower due to interactions with the SiO₂ surface – higher concentration implies stronger competition for the cation to get to the surface and stay adsorbed to it for a long time. Eventually, it has been shown that rotational motion of EMIM cations near the surface is somewhat slower than in bulk, but the effect is not large. However, it has turned out that to reproduce the relaxation data for the highest concentration (IGL-3) it was necessary to assume a larger (by about factor 2) relaxation dipole–dipole constant. This can suggest clustering effects due to the constraints imposed by the confinement.

Conflicts of interest

There are no conflicts of interest to declare.

Appendix

Comparison of the dependencies of ¹H spin relaxation data for EMIM-SCN in bulk and in SiO₂ confinement on temperature.

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